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The product of two negative numbers is 160. If the lesser of the two numbers is 4 less than twice the greater, what is the greater number?

A. –20

B. –16

C. –10

D. – 8

E. – 4

Algebra Second-Degree Equations

Let x and y be the two numbers, where x is the lesser of the two numbers and y is the number desired. From the given information, it follows that $xy = 160$ and $x = 2y - 4$, from which it follows that $(2y - 4)y = 160$. Dividing both sides of the last equation by 2 gives $(y - 2)y = 80$. Thus, 80 is to be written as a product of two negative numbers, one that is 2 less than the other. Trying simple factorizations of 80 quickly leads to the value of y : $(-40)(-2) = 80$, (–20)(–4) = 80, (–10)(–8) = 80. Therefore, $y = -8$. Note that because –8 is one of the answer choices, it is not necessary to ensure there are no other negative solutions to the equation $(y - 2)y = 80$.

Alternatively, $(y - 2)y = 80$ can be written as $y^2 - 2y - 80 = 0$. Factoring the left side gives $(y + 8)(y - 10) = 0$, and $y = -8$ is the only negative solution.

The correct answer is D.

•

According to a certain estimate, the depth $N(t)$, in centimeters, of the water in a certain tank at t hours past 2:00 in the morning is given by $N\left(t\right) = -20(t-5)^2 + 500$ for $0 \le t \le 10$. According to this estimate, at what time in the morning does the depth of the water in the tank reach its maximum?

A. 5:30

B. 7:00

C. 7:30

D. 8:00

E. 9:00

Algebra Functions

When $t = 5$, the value of $-20(t - 5)^2 + 500$ is 500. For all values of t between 0 and 10, inclusive, except $t = 5$, the value of $-20(t - 5)^2$ is negative and $-20(t - 5)^2 + 500 < 500$. Therefore, the tank reaches its maximum depth 5 hours after 2:00 in the morning, which is 7:00 in the morning.

The correct answer is B.

•

After driving to a riverfront parking lot, Bob plans to run south along the river, turn around, and return to the parking lot, running north along the same path. After running 3.25 miles south, he decides to run for only 50 minutes more. If Bob runs at a constant rate of 8 minutes per mile, how many miles farther south can he run and still be able to return to the parking lot in 50 minutes?

A. 1.5

B. 2.25

C. 3.0

D. 3.25

E. 4.75

Algebra Applied Problems

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After running 3.25 miles south, Bob has been running for $(3.25 \text{ miles})\left(8\frac{\text{minutes}}{\text{mile}}\right) = 26$ minutes. Thus, if t is the number of additional minutes that Bob can run south before turning around, then the number of minutes that Bob will run north, after turning around, will be $t + 26$. Since Bob will be running a total of 50 minutes after the initial 26 minutes of running, it follows that $t + (t + 26) = 50$, or $t = 12$. Therefore, Bob can run south an additional $\frac{12 \text{ minutes}}{8\frac{\text{minutes}}{\text{mile}}} = 1.5$ miles before turning around.

The correct answer is A.

•

Alex deposited x dollars into a new account that earned 8 percent annual interest, compounded annually. One year later Alex deposited an additional x dollars into the account. If there were no other transactions and if the account contained w dollars at the end of two years, which of the following expresses x in terms of w?

A. $\frac{w}{1 + 1.08}$

B. $\frac{w}{1.08 + 1.16}$

C. $\frac{w}{1.16 + 1.24}$

D. $\frac{w}{1.08 + (1.08)^2}$

E. $\frac{w}{(1.08)^2 + (1.08)^3}$

Algebra Applied Problems

At the end of the first year, the value of Alex's initial investment was $x(1.08)$ dollars, and after he deposited an additional x dollars into the account, its value was $[x(1.08) + x]$ dollars. At the end of the second year, the value was w dollars, where $w = \left[x\left(1.08\right) + x\right]\left(1.08\right) = x(1.08)^2 + x\left(1.08\right) = x\left[(1.08)^2 + 1.08\right]$. Thus, $x = \frac{w}{1.08 + (1.08)^2}$.

The correct answer is D.

•

M is the sum of the reciprocals of the consecutive integers from 201 to 300, inclusive. Which of the following is true?

A. $\frac{1}{3} < M < \frac{1}{2}$

B. $\frac{1}{5} < M < \frac{1}{3}$

C. $\frac{1}{7} < M < \frac{1}{5}$

D. $\frac{1}{9} < M < \frac{1}{7}$

E. $\frac{1}{12} < M < \frac{1}{9}$

Arithmetic Estimation

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M	$= \frac{1}{201} + \frac{1}{202} + \frac{1}{203} + \ldots + \frac{1}{298} + \frac{1}{299} + \frac{1}{300}$	Given
	$> \frac{1}{300} + \frac{1}{300} + \frac{1}{300} + \ldots + \frac{1}{300} + \frac{1}{300} + \frac{1}{300}$	See Note 1 below

Note 1: 300 is greater than each of 201, 202, 203, …, 298, and 299, so the reciprocal of 300 is less than the reciprocal of each of 201, 202, 203, …, 298, and 299 and the sum in line 2 of the table above, $(100)\left(\frac{1}{300}\right) = \frac{1}{3}$, is less than the sum of line 1. Thus, $\frac{1}{3} < M$.

M	$= \frac{1}{201} + \frac{1}{202} + \frac{1}{203} + \ldots + \frac{1}{298} + \frac{1}{299} + \frac{1}{300}$	Given
	$< \frac{1}{200} + \frac{1}{200} + \frac{1}{200} + \ldots + \frac{1}{200} + \frac{1}{200} + \frac{1}{200}$	See Note 2 below

Note 2: 200 is less than each of 201, 202, 203, …, 298, and 299, so the reciprocal of 200 is greater than the reciprocal of each of 201, 202, 203, …, 298, and 299 and the sum in line 2 of the table above, $\left(100\right) = \left(\frac{1}{200}\right) = \frac{1}{2}$, is greater than the sum of line 1. Thus, $\frac{1}{2} > M$ or $\frac{1}{2} < M$. Combining the results $\frac{1}{3} < M$ and $M < \frac{1}{2}$ gives $\frac{1}{3} < M < \frac{1}{2}$.

The correct answer is A.

•

Working simultaneously at their respective constant rates, Machines A and B produce 800 nails in x hours. Working alone at its constant rate, Machine A produces 800 nails in y hours. In terms of x and y, how many hours does it take Machine B, working alone at its constant rate, to produce 800 nails?

A. $\frac{x}{x+y}$

B. $\frac{y}{x+y}$

C. $\frac{xy}{x+y}$

D. $\frac{xy}{x-y}$

E. $\frac{xy}{y-x}$

Algebra Applied Problems

Let R_A and R_B be the constant rates, in nails per hour, at which Machines A and B work, respectively. Then it follows from the given information that $R_A + R_B = \frac{800}{x}$ and $R_A = \frac{800}{y}$. Hence, $\frac{800}{y} + R_B = \frac{800}{x}$, or $R_B = \frac{800}{x} - \frac{800}{y} = 800\left(\frac{1}{x} - \frac{1}{y}\right) = 800\left(\frac{y-x}{xy}\right)$. Therefore, the time, in hours, it would take Machine B to produce 800 nails is given by $\frac{800}{800} = \frac{xy}{y-x}$.

The correct answer is E.

•

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•

Carol purchased one basket of fruit consisting of 4 apples and 2 oranges and another basket of fruit consisting of 3 apples and 5 oranges. Carol is to select one piece of fruit at random from each of the two baskets. What is the probability that one of the two pieces of fruit selected will be an apple and the other will be an orange?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{24}$

D. $\frac{5}{24}$

E. $\frac{13}{24}$

Arithmetic Probability

There are 6 pieces of fruit in the first basket and 8 pieces of fruit in the second basket. By the multiplication principle, the number of selections of a piece of fruit from the first basket and a piece of fruit from the second basket is (6)(8) = 48. Of these selections, (4)(5) = 20 are such that an apple is selected from the first basket and an orange is selected from the second basket, and (2)(3) = 6 are such that an orange is selected from the first basket and an apple is selected from the second basket. Therefore, there is a total of 20 + 6 = 26 selections in which one apple is chosen and one orange is chosen, and hence the probability of such a selection is $\frac{26}{48} = \frac{13}{24}$.

Alternatively, the desired probability is the sum of the probabilities of two disjoint events. In the first event, an apple is selected from the first basket and an orange is selected from the second basket; the probability of this event is $\left(\frac{4}{6}\right)\left(\frac{5}{8}\right) = \frac{20}{48}$. In the second event, an orange is selected from the first basket and an apple is selected from the second basket; the probability of this event is $\left(\frac{2}{6}\right)\left(\frac{3}{8}\right) = \frac{6}{48}$. Therefore, the desired probability is

$\frac{20}{48} + \frac{6}{48} = \frac{26}{48} = \frac{13}{24}$.

The correct answer is E.

•

Last year Brand X shoes were sold by dealers in 403 different regions worldwide, with an average (arithmetic mean) of 98 dealers per region. If last year these dealers sold an average of 2,488 pairs of Brand X shoes per dealer, which of the following is closest to the total number of pairs of Brand X shoes sold last year by the dealers worldwide?

A. 10^4

B. 10^5

C. 10^6

D. 10^7

E. 10^8

Arithmetic Estimation; Exponents

Since the average number of dealers per region is 98, it follows that $\frac{\text{number of dealers}}{403} = 98$, and thus the number of dealers is (98)(403). Also, since the average number of pairs sold per dealer is 2,488, it follows that $\frac{\text{number of pairs sold}}{(98)(403)} = 2,488$, and thus the number of pairs sold is (98)(403)(2,488), which is approximately

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(100)(400)(2500) = $(10^2)(4 \times 10^2)(25 \times 10^2)$. Therefore, the total number of pairs sold is approximately $(4)(25) \times 10^{2+2+2} = 100 \times 10^6 = 10^8$.

The correct answer is E.

•

A positive integer is divisible by 3 if and only if the sum of its digits is divisible by 3. If the six-digit integer n is divisible by 3 and n is of the form $1k2, k24$, where k represents a digit that occurs twice, how many values could n have?

A. Two

B. Three

C. Four

D. Five

E. Ten

Arithmetic Equations

Since n is divisible by 3, the six digits of n sum to a number divisible by 3. So, $1 + k + 2 + k + 2 + 4 = 9 + 2k$ must be divisible by 3. The first term of $9 + 2k$ is 9, which is divisible by 3, so the other term $2k$ must be either 0 or else itself divisible by 3. That means k is either 0 or divisible by 3. Thus, since k is a single digit, it must be either 0, 3, 6, or 9. Each of these four values of k generates a different value for n: 102,024, 132,324, 162,624, or 192,924. Since no other value of k generates a six-digit number of the form $1k2, k24$ divisible by 3, but n must be a number of this form divisible by 3, these are the only four possible values of n.

The correct answer is C.

	Number of Marbles in Each of Three Bags	Percent of Marbles in Each Bag That Are Blue (to the nearest tenth)
Bag P	37	10.8%
Bag Q	x	66.7%
Bag R	32	50.0%

•

If $\frac{1}{3}$ of the total number of marbles in the three bags listed in the table above are blue, how many marbles are there in Bag Q?

A. 5

B. 9

C. 12

D. 23

E. 46

Algebra Percents

What is the value of x, the number of marbles in Bag Q? From the given information and rounded to the nearest integer, Bag P has (37)(0.108) = 4 blue marbles, Bag Q has $\left(x\right)\left(0.667\right) = \frac{2}{3}x$ blue marbles, and Bag R has (32)(0.5) = 16 blue marbles. Therefore, the total number of blue marbles is equal to $4 + \frac{2}{3}x + 16 = 20 + \frac{2}{3}x$. It is given that $\frac{1}{3}$ of the total number of marbles are blue, so the total number of blue marbles is also equal to $\frac{1}{3}\left(37 + x + 32\right) = \frac{1}{3}x + 23$. It follows that $20 + \frac{2}{3}x = \frac{1}{3}x + 23$, or $\frac{1}{3}x = 3$, or $x = 9$.

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The correct answer is B.

Age Category (in years)	Number of Employees
Less than 20	29
20–29	58
30–39	36
40–49	21
50–59	10
60–69	5
70 and over	2

•

The table above gives the age categories of the 161 employees at Company X and the number of employees in each category. According to the table, if m is the median age, in years, of the employees at Company X, then m must satisfy which of the following?

A. $20 \leq m \leq 29$

B. $25 \leq m \leq 34$

C. $30 \leq m \leq 39$

D. $35 \leq m \leq 44$

E. $40 \leq m \leq 49$

Arithmetic Statistics

The median of 161 ages is the 81st age when the ages are listed in order. Since 29 of the ages are less than 20, the median age must be greater than or equal to 20. Since 58 of the ages are between 20 and 29, a total of 29 + 58 = 87 of the ages are less than or equal to 29, and thus the median age is less than or equal to 29. Therefore, the median age is greater than or equal to 20 and less than or equal to 29.

The correct answer is A.

•

If k and n are positive integers such that $n > k$, then $k! + (n-k) \cdot (k-1)!$ is equivalent to which of the following?

A. $k \cdot n!$

B. $k! \cdot n$

C. $(n-k)!$

D. $n \cdot (k+1)!$

E. $n \cdot (k-1)!$

Algebra Simplifying Algebraic Expressions

$k! + (n-k) \cdot (k-1)!$	given
$k \cdot (k-1)! + (n-k) \cdot (k-1)!$	$k! = k \cdot (k-1)!$
$[k + (n-k)] \cdot (k-1)!$	factor out $(k-1)!$
$n \cdot (k-1)!$	combine like terms

The correct answer is E.

•

Ron is 4 inches taller than Amy, and Barbara is 1 inch taller than Ron. If Barbara's height is 65 inches, what is the median height, in inches, of these three people?

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median height, in inches, of these three people?

A. 60

B. 61

C. 62

D. 63

E. 64

Arithmetic Operations with Integers

Let R, A, and B be the heights, respectively and in inches, of Ron, Amy, and Barbara. It is given that $R = 4 + A$, $B = 1 + R$, and $B = 65$. Therefore, $R = B - 1 = 65 - 1 = 64$ and $A = R - 4 = 64 - 4 = 60$. From this it follows that the three heights, in inches, are 60, 64, and 65. The median of these three heights is 64.

The correct answer is E.

•

If x and y are positive numbers such that $x + y = 1$, which of the following could be the value of $100x + 200y$?

I. 80

II. 140

III. 199

A. II only

B. III only

C. I and II

D. I and III

E. II and III

Algebra Simultaneous Equations; Inequalities

Since $x + y = 1$, then $y = 1 - x$ and $100x + 200y$ can be expressed as $100x + 200(1 - x) = 200 - 100x$.Test each value.

I. If $200 - 100x = 80$, then $x = \frac{200 - 80}{100} = 1.2$ and $y = 1 - 1.2 = -0.2$.

Since y must be positive, 80 cannot be a value of $100x + 200y$.

II. If $200 - 100x = 140$, then $x = \frac{200 - 140}{100} = 0.6$ and $y = 1 - 0.6 = 0.4$, so 140 can be a value of $100x + 200y$.

III. If $200 - 100x = 199$, then $x = \frac{200 - 199}{100} = 0.01$ and $y = 1 - 0.01 = 0.99$, so 199 can be a value of $100x + 200y$.

The correct answer is E.

•

If X is the hundredths digit in the decimal $0.1X$ and if Y is the thousandths digit in the decimal $0.02Y$, where X and Y are nonzero digits, which of the following is closest to the greatest possible value of $\frac{0.1X}{0.02Y}$?

A. 4

B. 5

C. 6

D. 9

E. 10

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E. 10

Arithmetic Operations with Decimals; Place Value

The greatest possible value of $\frac{0.1X}{0.02Y}$ will occur when $0.1X$ has the greatest possible value and $0.02Y$ has the least possible value. Since X and Y are nonzero digits, this means that X must be 9 and Y must be 1. The greatest possible value of $\frac{0.1X}{0.02Y}$ is then $\frac{0.19}{0.021} \approx 9.05$, which is closest to 9.

The correct answer is D.

•

If each of the 12 teams participating in a certain tournament plays exactly one game with each of the other teams, how many games will be played?

A. 144

B. 132

C. 66

D. 33

E. 23

Arithmetic Elementary Combinatorics

Since each of the 12 teams will play exactly one game with each of the other teams, the number of games that will be played is equal to the number of selections of 2 teams, without regard to order, from 12 teams. This is the number of combinations of 12 teams taken 2 at a time, which is equal to

$$\binom{12}{2} = \frac{12!}{2!\,(12-2)!} = \frac{\cancel{(10!)}\ (11)\,(12)}{(2)} = 66.$$

Alternatively, each of the 12 teams will play each of the 11 other teams. The product (12)(11) counts each of the games twice since, for example, this product separately counts "Team A plays Team B" and "Team B plays Team A." Therefore, the number of games that will be played is $\frac{(12)\,(11)}{2} = 66.$

The correct answer is C.

•

For which of the following values of x is $\sqrt{1-\sqrt{2-\sqrt{x}}}$ not defined as a real number?

A. 1

B. 2

C. 3

D. 4

E. 5

Algebra Equations

For the difference between two numbers to be a real number, those two numbers must both be real themselves. So, for $\sqrt{1-\sqrt{2-\sqrt{x}}}$ to be a real number, $\sqrt{2-\sqrt{x}}$ must be a real number. Thus, since only a nonnegative real number has a real square root, $2-\sqrt{x}$ must be a nonnegative real number. It follows that $\sqrt{x}$ must be a real number no greater than 2, so x must be a real number no greater than 4. Therefore, for $x = 5$, $\sqrt{1-\sqrt{2-\sqrt{x}}}$ is not defined as a real number.

The correct answer is E.

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•

What is the remainder when 3^{19} is divided by 10?

A. 1

B. 3

C. 5

D. 7

E. 9

Arithmetic **Properties of Integers; Series and Sequences**

When a positive integer is divided by 10, the remainder is the units digit of that positive integer. For example, when 584 is divided by 10, the quotient is 58 and the remainder is 4. Thus, the remainder when 3^{19} is divided by 10 is the units digit when 3^{19} is written as a decimal numeral. The table shows the units digits for the first few powers of 3.

number	decimal numeral	units digit
3^1	3	3
3^2	9	9
3^3	27	7
3^4	81	1
3^5	243	3
3^6	729	9
3^7	2,187	7
3^8	6,561	1
⋮	⋮	⋮

The table, along with thinking about the standard method for multiplying decimal numerals by hand, shows that the units digits of the powers of 3 form a sequence in which the digits 3, 9, 7, and 1 repeat indefinitely in that order. By following this pattern, the units digit of 3^{19} is seen to be 7.

The correct answer is D.

•

Of 200 people surveyed, 80 percent own a cellular phone and 45 percent own a pager. If all 200 people surveyed own a cellular phone, or a pager, or both, what percent of those surveyed either do not own a cellular phone or do not own a pager?

A. 35%

B. 45%

C. 55%

D. 65%

E. 75%

Algebra **Sets (Venn Diagrams)**

From the given information, (0.8)(200) = 160 of the people surveyed own a cellular phone and (0.45)(200) = 90 of the people surveyed own a pager. The Venn diagram shows this information, where N denotes the number of people who own both.

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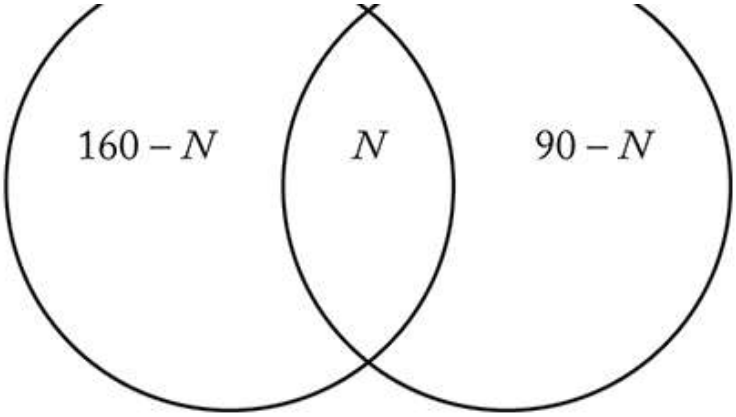

Since a total of 200 people were surveyed, it follows that $(160 - N) + N + (90 - N) = 200$, or $N = 50$. Therefore, the number of people surveyed who do not own a cellular phone is 90 − 50 = 40 and the number of people surveyed who do not own a pager is 160 − 50 = 110, which together account for 40 + 110 = 150 of the 200 people surveyed, or $\frac{150}{200} \times 100\% = 75\%$ of the people surveyed.

The correct answer is E.

•

At the end of the first quarter, the share price of a certain mutual fund was 20 percent higher than it was at the beginning of the year. At the end of the second quarter, the share price was 50 percent higher than it was at the beginning of the year. What was the percent increase in the share price from the end of the first quarter to the end of the second quarter?

A. 20%

B. 25%

C. 30%

D. 33%

E. 40%

Arithmetic Applied Problems; Percents

Let P be the sale price of the mutual fund at the beginning of the year. Then at the end of the first quarter the sale price was $1.2P$, and at the end of the second quarter the sale price was $1.5P$. Therefore, the percent increase in the sale price from the end of the first quarter to the end of the second quarter is $\frac{1.5P - 1.2P}{1.2P} \times 100\% = \frac{15 - 12}{12} \times 100\% = 25\%$.

The correct answer is B.

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots$$

•

In the sequence above, each term after the first is one-half the previous term. If x is the tenth term of the sequence, then x satisfies which of the following inequalities?

A. $0.1 < x < 1$

B. $0.01 < x < 0.1$

C. $0.001 < x < 0.01$

D. $0.0001 < x < 0.001$

E. $0.00001 < x < 0.0001$

Arithmetic Series and Sequences; Decimals

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The first, second, third, fourth, etc. terms are, respectively, the reciprocals of 2^1, 2^2, 2^3, 2^4, etc. Therefore, the tenth term x is the reciprocal of

2^{10} = 1,024, or $\frac{1}{1,024}$. Since 1,000 < 1,024 < 10,000,

it follows that $\frac{1}{10,000} < \frac{1}{1,024} < \frac{1}{1,000}$, so $0.0001 < \frac{1}{1,024} < 0.001$, or $0.0001 < x < 0.001$.

The correct answer is D.

•

If $xy + z = x(y + z)$, which of the following must be true?

A. $x = 0$ and $z = 0$

B. $x = 1$ and $y = 1$

C. $y = 1$ and $z = 0$

D. $x = 1$ or $y = 0$

E. $x = 1$ or $z = 0$

Algebra Equations

$xy + z = x(y + z)$	given
$xy + z = xy + xz$	distributive property
$z = xz$	subtract xy from both sides
$0 = xz - z$	subtract z from both sides
$0 = (x - 1)z$	factor
$x - 1 = 0$ or $z = 0$	property of 0
$x = 1$ or $z = 0$	solve for x

The correct answer is E.

•

Two water pumps, working simultaneously at their respective constant rates, took exactly 4 hours to fill a certain swimming pool. If the constant rate of one pump was 1.5 times the constant rate of the other, how many hours would it have taken the faster pump to fill the pool if it had worked alone at its constant rate?

A. 5

B. $\frac{16}{3}$

C. $\frac{11}{2}$

D. 6

E. $\frac{20}{3}$

Arithmetic Work Problem

Let t be the number of hours it would take the slower pump to fill the pool working alone. Since the faster pump works $1.5 = \frac{3}{2}$ times faster than the slower pump, the faster pump will take $\frac{1}{1.5}t = \frac{2}{3}t$ hours to fill the pool working alone. It follows that $\frac{1}{t} + \frac{1}{\frac{2}{3}} = \frac{1}{4}$, or $\left(1 + \frac{3}{2}\right)\left(\frac{1}{t}\right) = \frac{1}{4}$.

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Solving, we get $\frac{1}{t} = \frac{1}{4} \div \frac{5}{2} = \frac{1}{10}$, and hence $t = 10$.

Therefore, the faster pump would have taken $\frac{2}{3}t = \frac{2}{3} \times 10 = \frac{20}{3}$ hours to fill the pool if it had worked alone.

Alternatively, let r and $1.5r = \frac{3}{2}r$ be the rates, respectively, at which the slower and faster pumps work. Working together, the pumps work at a combined rate of $r + \frac{3}{2}r = \frac{5}{2}r$ and can fill 1 pool in 4 hours. Therefore, when working alone, the faster pump, which works at a rate that is $\frac{3}{5}$ times as fast as their combined rate (because $\frac{3}{2}r = \frac{3}{5} \times \frac{5}{2}r$), will take $\frac{1}{\frac{3}{5}} = \frac{5}{3}$ times as long as than when they are working together, or $\frac{5}{3} \times 4 = \frac{20}{3}$ hours to fill the pool.

The correct answer is E.

•

If $0 < x < 1$, what is the median of the values x, x^{-1}, x^2, $\sqrt{x}$, and x^3?

A. x

B. x^{-1}

C. x^2

D. $\sqrt{x}$

E. x^3

Arithmetic Order

Since we can assume the correct answer choice does not depend on the value of x between 0 and 1 that is used, choose a value of x between 0 and 1 for which all of these expressions can be easily evaluated, such as $x = \frac{1}{4}$.

x	x^{-1}	x^2	$\sqrt{x}$	x^3
$\frac{1}{4}$	4	$\frac{1}{16}$	$\frac{1}{2}$	$\frac{1}{64}$

When these values are arranged in ascending order,

$$\frac{1}{64} \quad \frac{1}{16} \quad \frac{1}{4} \quad \frac{1}{2} \quad 4$$

the median of the values will be the middle value, which is $x = \frac{1}{4}$.

The correct answer is A.

•

The number of stamps that Kaye and Alberto had were in the ratio 5:3, respectively. After Kaye gave Alberto 10 of her stamps, the ratio of the number Kaye had to the number Alberto had was 7:5. As a result of this gift, Kaye had how many more stamps than Alberto?

A. 20

B. 30

C. 40

D. 60

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E. 90

Algebra Ratio and Proportion

Let x be the number of stamps Alberto had at first. Then Kaye had $\frac{5x}{3}$ stamps at first. So, after Kaye gave Alberto 10 of her stamps, she had $\frac{5x}{3} - 10$ stamps and Alberto had $x + 10$ stamps. Since Kaye then had 7 stamps for every 5 Alberto had, $7(x + 10) = 5\left(\frac{5x}{3} - 10\right) = 7x + 70 = \frac{25x}{3} - 50$. Thus, $120 = \frac{(25 - 21)x}{3} = \frac{4x}{3}$, so $360 = 4x$, and $x = 90$. Therefore, after Kaye gave Alberto 10 of her stamps, she had $\frac{5\,(90)}{3} - 10 = 5\left(30\right) - 10 = 140$ stamps, while Alberto had 90 + 10 = 100 stamps. So, as a result of the gift, Kaye had 140 − 100 = 40 more stamps than Alberto.

The correct answer is C.

7.51	8.22	7.86	8.36
8.09	7.83	8.30	8.01
7.73	8.25	7.96	8.53

•

A vending machine is designed to dispense 8 ounces of coffee into a cup. After a test that recorded the number of ounces of coffee in each of 1,000 cups dispensed by the vending machine, the 12 listed amounts, in ounces, were selected from the data. If the 1,000 recorded amounts have a mean of 8.1 ounces and a standard deviation of 0.3 ounce, how many of the 12 listed amounts are within 1.5 standard deviations of the mean?

A. Four

B. Six

C. Nine

D. Ten

E. Eleven

Arithmetic Statistics

Since 1.5 standard deviations is equal to (1.5)(0.3) = 0.45, it follows that a recorded number of ounces of coffee is within 1.5 standard deviations of the mean if that number of ounces is within 0.45 of 8.1, or between 8.1 − 0.45 = 7.65 and 8.1 + 0.45 = 8.55. Since 7.51 is the only one of the 12 listed amounts that is not between 7.65 and 8.55, there are 12 − 1 = 11 listed amounts that are between 7.65 and 8.55.

The correct answer is E.

•

In a certain quiz that consists of 10 questions, each question after the first is worth 4 points more than the preceding question. If the 10 questions on the quiz are worth a total of 360 points, how many points is the third question worth?

A. 18

B. 24

C. 26

D. 32

E. 44

Algebra Series and Sequences

Suppose the first question on the quiz is worth x points. Then the 10 questions together are worth $10x + 4(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9) = 10x + 4(45) = 10x + 180 = 360$ points. So, $10x = 180$, and $x = 18$. Since the third question is worth 8 points more than the first, it must be worth 18 + 8 = 26 points.

The correct answer is C.

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The correct answer is C.

•

A car traveling at a certain constant speed takes 2 seconds longer to travel 1 kilometer than it would take to travel 1 kilometer at 75 kilometers per hour. At what speed, in kilometers per hour, is the car traveling?

A. 71.5

B. 72

C. 72.5

D. 73

E. 73.5

Algebra Applied Problems

An hour is 3,600 seconds. At 75 kilometers per hour, the car would take 3,600 seconds to travel 75 kilometers, so it would take $\frac{3600}{75} = 48$ seconds to travel 1 kilometer. But we're told that the car actually takes 2 seconds longer than that, or 50 seconds, to travel 1 kilometer. Therefore, in 1 hour, or 3,600 seconds, it travels $\frac{3600}{50} \times 1$ kilometer/hour = 72 kilometers/hour.

The correct answer is B.

•

If the speed of x meters per second is equivalent to the speed of y kilometers per hour, what is y in terms of x? (1 kilometer = 1,000 meters)

A. $\frac{5x}{18}$

B. $\frac{6x}{5}$

C. $\frac{18x}{5}$

D. $60x$

E. $3,600,000x$

Algebra Measurement Conversion

We are given that y kilometers per hour is the same speed as x meters per second. By converting from meters to kilometers using 1 km = 1,000 m and seconds to hours using 1 hr = 3,600 sec, we can obtain y in terms of x.

$$y\frac{\text{km}}{\text{hr}} = x\frac{\text{m}}{\text{sec}}$$

$$y\frac{\text{km}}{\text{hr}} = x\frac{\cancel{\text{m}}}{\cancel{\text{sec}}} \times \frac{1\text{ km}}{1,000\cancel{\text{m}}} \times \frac{3,600\cancel{\text{sec}}}{1\text{ hr}}$$

$$y\frac{\text{km}}{\text{hr}} = \left(\frac{3,600}{1,000}\right)x\frac{\text{km}}{\text{hr}}$$

$$y\frac{\text{km}}{\text{hr}} = \frac{18x}{5}\frac{\text{km}}{\text{hr}}$$

From the last equality above it follows that $y = \frac{18x}{5}$.

The correct answer is C.

•

The function f is defined by $f\left(x\right) = -\frac{1}{x}$ for all nonzero numbers x. If $f\left(a\right) = -\frac{1}{2}$ and $f\left(ab\right) = \frac{1}{6}$, then

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$b =$

A. 3

B. $\frac{1}{3}$

C. $-\frac{1}{x}$

D. –3

E. –12

Algebra Functions

From $f\left(x\right) = -\frac{1}{x}$ and $f\left(a\right) = -\frac{1}{2}$ we have $-\frac{1}{a} = -\frac{1}{2}$,

or $a = 2$. From $f\left(x\right) = -\frac{1}{x}$ and $f\left(ab\right) = \frac{1}{6}$ we have $-\frac{1}{ab} = \frac{1}{6}$, or $ab = -6$. Since $a = 2$, it follows from

$ab = -6$ that $2b = -6$, or $b = -3$.

The correct answer is D.

•

If $5^x - 5^{x-3} = (124)(5^y)$, what is y in terms of x?

A. x

B. $x - 6$

C. $x - 3$

D. $2x + 3$

E. $2x + 6$

Algebra Equations; Factoring

A strategy that is useful in rewriting sums and/or differences of exponential terms having the same base is to use the term with the least exponent as a common factor. For example, in $5^x - 5^{x-3}$, factor out 5^{x-3} since $x - 3$ is less than x. This gives $5^x - 5^{x-3} = 5^{x-3}(5^3 - 1)$. Therefore, the given equation is equivalent to $5^{x-3}(5^3 - 1) = (124)(5^y)$, or $5^{x-3}(124) = (124)(5^y)$. Dividing both sides of the last equation by 124 gives $5^{x-3} = 5^y$, and hence $y = x - 3$.

The correct answer is C.

•

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